

Aditya Kumar

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SUMMARY

MS Aeronautics and Astronautics student focused on GPU-accelerated CFD, solver verification, and surrogate modeling across production HPC codes (CUDA, HIP, MPI, Fortran), commercial CFD (ANSYS Fluent, STAR-CCM+, OpenFOAM), and ML surrogates.

EDUCATION

University of Washington

MS Aeronautics and Astronautics (Fluids Concentration), GPA: 3.88/4.0

Seattle, WA

Sep 2025 - Present

Vishwakarma Institute of Technology

BTech Mechanical Engineering, CGPA: 8.42/10 (Top 10%)

Pune, India

Jul 2021 - May 2025

EXPERIENCE

Research Assistant, Computational Fluid Mechanics Lab

University of Washington

Seattle, WA

Sep 2025 - Present

- Porting the pressure-Poisson solver of a production DNS code (C/CUDA, Fortran, MPI) from NVIDIA to AMD GPUs using HIP and hipFFT, verified against Taylor-Green vortex regressions on every change
- Optimized the same Poisson solve on NVIDIA architectures and automated the Python regression pipeline that gates the port, extending the lab's ptest random right-hand-side operator suite

Research Co-op, Thermal Systems Engineering

Kirloskar Pneumatic Co. Ltd.

Pune, India

Jul 2024 - May 2025

- Developed Python 1D thermodynamic models of twin-screw compressors against CoolProp, integrating six leakage paths to close a 66 g/cycle mass-flow gap against vendor software and improving compression efficiency by 2%
- Trained a Gaussian Process surrogate (POD dimensionality reduction, Matern kernel) on 800 to 1200 CFD cases generated via SCORG and Simerics, mapping rotor geometry directly to chamber-cycle response

Aerodynamics Engineer

Veloce Racing India (Formula Student)

Pune, India

Aug 2021 - Apr 2023

- Automated full-vehicle CFD workflows in ANSYS Fluent and STAR-CCM+ via the PyFluent Settings API, achieving 5% drag reduction through DOE-driven optimization across 30+ configurations
- Coordinated wind tunnel validation campaigns and mentored 5 junior engineers on CFD best practices; team finished 1st nationally and 2nd globally in the combustion category

Manufacturing Engineering Intern

Jaguar Land Rover

Pune, India

May 2023 - Aug 2023

- Optimized Oil Filling and Cotter Press machinery through design-for-manufacturing changes, cutting unplanned downtime 15%

PROJECTS

2D Incompressible Navier-Stokes Solver

Julia, FFT and CG Poisson solvers, CairoMakie

May 2026 - Present

- Built a 2D incompressible Navier-Stokes solver from scratch on a staggered MAC grid with RK3 fractional-step integration (Le and Moin, JCP 1991); full suite passes 15 of 15 tests in 5.2 s wall clock
- Verified second-order spatial accuracy by method of manufactured solutions (L2 slopes 1.994, 1.999, 2.000 across 16 to 128 point grids) and matched Ghia et al. (1982) cavity centerline profiles at Re 100 and Re 400

Conjugate Heat Transfer Study, Liquid-Cooled Cold Plate

OpenFOAM, Python, scikit-learn, SLURM, Apptainer

Apr 2026 - Aug 2026

- Automated a 96-case conjugate heat transfer sweep of a microchannel cold plate under a 700 W GPU-class heat load via a templated SLURM array pipeline on an HPC cluster; 89 of 96 cases converged
- Trained Gaussian Process surrogates and ran constrained optimization to cut thermal resistance 22% at matched pumping power; validated against Shah-London (0.37% error) and a Roache GCI study (2.21% pressure drop)

TECHNICAL SKILLS

Programming: Python, C, C++, CUDA, HIP, Fortran, Julia, MATLAB, Bash/Shell, SQL

HPC and Parallel Computing: MPI, OpenMP, GPU Computing (CUDA, HIP, cuFFT, hipFFT), SLURM, PBS

CFD and Simulation: ANSYS Fluent, PyAnsys (PyFluent), STAR-CCM+, OpenFOAM, ParaView

ML and Data Tools: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Streamlit

Numerical Methods: Finite Volume Discretization, RANS/LES Turbulence Modeling, DOE, ROM, PINNs